$$r(t) = \prod_i \left\langle \hat{D}\left(\lambda_i(t)\right) \right\rangle_{\hat{\rho}_{\mathcal{E}_i}} = \prod_i \left\langle \exp\left[\lambda_i(t)\hat{a}_i^\dagger - \lambda_i^*(t)\hat{a}_i\right] \right\rangle_{\hat{\rho}_{\mathcal{E}_i}}, \qquad (5.118)$$